RUNGTA INTERNATIONAL SKILLS UNIVERSITY

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SCHOOL OF INFORMATION TECHNOLOGY

BACHELOR OF COMPUTER APPLICATION

PS No.	**86**
Project Title	**Retail Customer Analyzer**

Problem Statement	Read customers.csv with Pandas. Show mean, median, highest and lowest spending, list customers who spent at least a limit entered by the user, and show city-wise total and average spending.
	Language: Python 3 **Concepts:** Pandas, Filtering, Grouping, Statistics
	Sample data (5 columns, 8 rows):
	`CustomerID,Name,City,Purchases,TotalSpent`
	`C01,Asha,Raipur,5,12000`
	`C02,Bhavin,Durg,2,3000`
	`C03,Chitra,Raipur,8,25000`
	`C04,Deepak,Bhilai,1,800`
	`C05,Esha,Durg,4,9000`
	`C06,Farhan,Raipur,3,6000`
	`C07,Gita,Bhilai,6,15000`
	`C08,Hari,Durg,2,2200`

Test Cases / Case Statements (4 cases)	**INPUT** / **EXPECTED OUTPUT** **Test Case 1: Normal Input** File: customers.csv (sample above) Menu: Spending statistics Mean spend: Rs. 9125.00 Median spend: Rs. 7500.00 Highest: Chitra (Rs. 25000) Lowest: Deepak (Rs. 800) **Test Case 2: Boundary / Minimum Input** Menu: High-value customers Limit: 25000 High-value customers (TotalSpent >= 25000): 1 C03 | Chitra | Raipur | Rs. 25000 **Test Case 3: Invalid Input** (a) Menu: High-value customers, Limit: abc (b) Menu: High-value customers, Limit: -500 (a) Invalid amount. Please enter a number. (b) Amount must be 0 or more. **Test Case 4: Different Valid Scenario** File: customers.csv (sample above) Menu: City-wise summary Raipur : Total Rs. 43000 | Average Rs. 14333.33 Bhilai : Total Rs. 15800 | Average Rs. 7900.00 Durg : Total Rs. 14200 | Average Rs. 4733.33
Bare Minimum Requirements	1. Write the solution in Python 3 (one .py file) and run it from the console. 2. Load customers.csv with Pandas and handle a missing file. 3. Calculate mean, median, highest and lowest TotalSpent. 4. Filter and list customers whose TotalSpent is at least the limit entered by the user. 5. Test the program with all four test cases.
Instruction	1. Understand the problem statement and define the problem clearly. 2. Develop the solution using proper logic and programming concepts. 3. Implement all required features and pass the given test cases. 4. Test and debug the project to ensure correct output. 5. Prepare the documentation/report and upload the code and report to GitHub. 6. Complete and submit the project within the given deadline.